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Original Article

Neural representations of beat and rhythm in motor and association regions

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Humans perceive a pulse, or beat, underlying musical rhythm. Beat strength correlates with activity in the basal ganglia and supplementary motor area, suggesting these regions support beat perception. However, the basal ganglia and supplementary motor area are part of a general rhythm and timing network (regardless of the beat) and may also represent basic rhythmic features (e.g. tempo, number of onsets). To characterize the encoding of beat-related and other basic rhythmic features, we used representational similarity analysis. During functional magnetic resonance imaging, participants heard 12 rhythms—4 strong-beat, 4 weak-beat, and 4 nonbeat. Multi-voxel activity patterns for each rhythm were tested to determine which brain areas were beat-sensitive: those in which activity patterns showed greater dissimilarities between rhythms of different beat strength than between rhythms of similar beat strength. Indeed, putamen and supplementary motor area activity patterns were significantly dissimilar for strong-beat and nonbeat conditions. Next, we tested whether basic rhythmic features or models of beat strength (counterevidence scores) predicted activity patterns. We found again that activity pattern dissimilarity in supplementary motor area and putamen correlated with beat strength models, not basic features. Beat strength models also correlated with activity pattern dissimilarities in the inferior frontal gyrus and inferior parietal lobe, though these regions encoded beat and rhythm simultaneously and were not driven by beat alone.

Key words: basal ganglia; fMRI; rhythm; Mvpa; SMA.

Introduction

Regular rhythms, in which onsets often align with an evenly timed metric "grid," elicit a strong sense of an underlying pulse or "beat" unfolding in time with the rhythm (Bouwer et al. 2018). Compared to irregular rhythms, which do not give rise to a beat, regular (or strong-beat) rhythms elicit greater activity in the supplementary motor area (SMA) and basal ganglia, specifically the putamen, bilaterally (Chen et al. 2006, 2008a; Grahn and Brett 2007; Bengtsson et al. 2009; Grahn and Rowe 2009, 2013; Chapin et al. 2010; Matthews et al. 2020; Hoddinott et al. 2021), suggesting these regions are involved in beat perception. The basal ganglia and SMA also activate during rhythms with no beat (but less than when a beat is present), suggesting that regional activity may relate to more than just beat, and other features of rhythm (such as tempo) may be represented. Representation of rhythmic features in beat-sensitive regions may be overlooked in previous studies due to the coarse nature of univariate analyses, in which activity is averaged across voxels in a region, and across stimuli in a condition, obscuring differences in rhythmic features between individual rhythms. Here, we employ more sensitive multivariate pattern analyses (MVPAs) to test whether SMA and basal ganglia activity represent only beat strength or if these regions are also sensitive to other individual rhythmic features. We also used these analyses to examine whether other regions showed evidence of representing the beat, as this evidence may have been obscured in univariate analyses.

The basal ganglia and SMA are key components in a striatal cortical loop involved in beat perception. During the presentation of rhythms with a strong beat, functional connectivity between the basal ganglia, SMA, and auditory sensory areas is increased (Grahn 2009), perhaps reflecting inter-regional communication of external auditory events (via auditory cortex) and internal beat-based predictions (via SMA and basal ganglia) (Cannon and Patel 2021). Of these regions, the basal ganglia are particularly crucial to beat perception, as patients with Parkinson's disease (in which basal ganglia function is compromised) show a selective deficit for strong-beat rhythms in discrimination tasks (Grahn and Brett 2009). The basal ganglia's role in beat perception may relate to prediction, as putamen activity is greatest while maintaining (i.e. predicting) an ongoing beat, compared to initially detecting a beat (Grahn and Rowe 2013). There is also strong evidence for beat sensitivity in the SMA, in which activity is greatest for rhythms with a strong beat compared to nonbeat rhythms (Grahn and Brett 2007; Grahn and Rowe 2009, 2013) and predicts subjective beat perception ability—individuals who easily detect a beat have greater beat-related SMA activity than individuals who find it more difficult to perceive a beat (Grahn and McAuley 2009; Grahn and Schuit 2012). The mechanistic role of the SMA in beat perception is less clear, though its involvement in temporal processing across nonrhythmic tasks suggests that the SMA may encode rhythm or temporal information in general, not just when a beat is present.

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The SMA and basal ganglia encode temporal features in nonrhythmic tasks and across sensory domains, suggesting that general temporal processing may drive activity during rhythm perception, in addition to beat-related activity. For example, SMA, caudate, and putamen activate during auditory and tactile temporal discrimination—detecting whether one or two events were presented sequentially (Pastor et al. 2004, 2006). Putamen activity during temporal discrimination corresponds to individual perceptual certainty—putamen activity is greatest when subjects are highly consistent in their responses (Pastor et al. 2008), perhaps reflecting the accuracy of temporal encoding during the task. Similarly, duration estimation tasks recruit the SMA and basal ganglia, in addition to inferior frontal gyri (IFGs) and insular cortices (Coull et al. 2015). Activity in the SMA increases as interval duration increases and corresponds to subjective overestimation of the intervals—greater SMA activity during initial encoding of the interval increased the likelihood for the duration to be overestimated behaviorally, suggesting that SMA activity relates to temporal interval encoding accuracy (Coull et al. 2015). Indeed, recent work has shown SMA activity encodes visually presented intervals in a "chronotopic map" such that different interval durations activate distinct topographic regions across the supplementary motor cortex (Protopapa et al. 2019), akin to the "tonotopic" maps of pitches in auditory cortex (Humphries et al. 2010). Using multi-interval sequences, the SMA and basal ganglia activate during encoding of visually presented sequences of temporal intervals (i.e. visual rhythms; Schubotz and von Cramon 2001), and the SMA exhibits distinct multivoxel activation patterns for the timing and order of learned finger-press sequences (Kornysheva and Diedrichsen 2014), suggesting that both the perception and production of multi-interval timing is represented in the SMA and basal ganglia. Thus, the activity in the SMA and basal ganglia during auditory rhythm perception is likely to (at least partially) represent the rhythms themselves—the multiduration intervals making up the rhythmic sequences—though this has not been shown when also controlling for beat strength.

Because the SMA and basal ganglia respond generally to temporal processing across tasks and modalities, these motor regions are likely responsible for encoding auditory rhythms, regardless of beat presence. However, in one auditory rhythm perception study, a linear classifier could not accurately discriminate between two rhythms based on SMA activity patterns (Notter et al. 2019) but could discriminate based on the left and right temporoparietal junction and the right IFG patterns. The SMA, though, was found to be part of a broad network that distinguished between multi-interval rhythms and an isochronous metronome. Notter et al. (2019) suggest that this network may reflect difficulty differences between rhythms and metronome, rather than multi-interval encoding, especially for regions not known to be involved in timing (e.g. the lateral occipital lobe). However, the SMA may represent a stimulus feature that is balanced across the rhythms but absent in the metronome, preventing classification between the rhythms but enabling it between rhythms and metronome. For example, the two rhythms had the same number of short, medium, and long intervals, and the metronome only had one repeating interval, so if the durations of multiple intervals were being encoded (regardless of their order), both rhythms would elicit identical activity. Similarly, both rhythms were metric and could elicit the internal generation of a beat, whereas metronomes do not require the internal generation of a beat. Thus, it is possible that the SMA (and perhaps other regions) encoded either the presence of a beat or basic features common to both rhythms, allowing a classifier to distinguish between rhythms and a metronome but not between the rhythms themselves. The SMA may encode individual rhythms, based on temporal features such as interval lengths or tempo, in addition to beat strength. Thus, more evidence is needed to characterize the neural representation of rhythm and beat to determine whether rhythm perception regions show different activity patterns based on beat strength or other basic rhythmic features or both.

One method to characterize neural representations in functional magnetic resonance imaging (fMRI) is representational similarity analysis (RSA). RSA operates under the assumption that neurons, and more widely neuronal populations, are tuned to specific stimulus features. For example, an auditory neuron may fire specifically for sound frequencies near 440 Hz but not other frequencies (Bendor and Wang 2005). To communicate information downstream, any "tuned" region must output activity that is consistent and sufficiently distinct to be decoded by the region receiving the input. Although fMRI has a coarser spatial resolution than single neuron recordings, it can still show differences in the tuning of populations of neurons to a task, stimulus, or a combination of stimulus features. Thus, RSA measures encoding by the relative dissimilarity (i.e. the "decodability") between activity patterns for stimuli that differ along the encoded feature (Kriegeskorte et al. 2008). For example, a region that encodes beat strength may exhibit dissimilar activity patterns across its voxels in response to rhythms that have a strong beat versus no beat but similar patterns for different rhythms with similar beat strength. Therefore, RSA can quantify dissimilarities between multivoxel activity patterns for individual rhythms and determine how stimulus features (such as beat strength) are represented across different brain regions.

Representational similarity analysis is a form of MVPA, which provides complementary information to traditional univariate analysis. In some cases, MVPA is more sensitive than univariate analysis. For example, when learning motor sequences, average regional activity may increase or decrease across training, while multivariate activity patterns for each motor sequence become more distinct from each other, suggesting that the neural representation of individual trained motor sequences is strengthened (Wiestler and Diedrichsen 2013). Univariate analysis is appropriate for identifying regions *involved* in a task; however, MVPA is better equipped to characterize *how* information is represented in an involved region (Mur et al. 2009). Although several neuroimaging studies have shown motor region involvement during rhythm and beat perception, to date, none have examined how rhythms are represented in regional activity patterns. For example, certain brain areas may be sensitive only to the presence or absence of a beat, or they may also encode additional rhythmic features.

In this experiment, we characterized the neural representations of rhythm and beat perception using RSA and high-resolution 7-Tesla fMRI. During MRI scanning, participants listened to a set of 12 unique rhythms taken from previous beat perception work (Grahn and Brett 2007). Of the 12 rhythms, 4 had a strong beat, 4 had a weak beat, and 4 had no beat. Activity patterns for each rhythm were extracted from the a priori areas of interest (SMA, basal ganglia), as well as the whole brain. We predicted general rhythm encoding in the SMA and basal ganglia—highly dissimilar multivariate patterns between all rhythms. If the SMA and basal ganglia are sensitive mainly to beat strength, we would predict greater dissimilarity between strong-beat and weak- and nonbeat rhythms, compared to within-condition dissimilarities (e.g. the dissimilarity between patterns for the 4 strong-beat rhythms). Across the whole brain, we predicted rhythm encoding in the rhythm perception network, including the IFG, premotor

Table 1. Rhythm list including interval ratios and absolute durations.

	Interval ratios	Base unit (ms)	Interval durations (ms)
Strong Beat	1 1 2 3 1 4	270	270,270,810,270,1080
	3 1 1 3 2 2	250	750,250,250,750,500,500
	2 1 1 3 1 1 3	230	460,230,230,690,230,230,690
	3 1 4 1 1 1 1	250	750,250,1000,250,250,250,250
Weak Beat	3 2 3 2 1 1	270	810,540,810,540,270,270
	4 1 2 2 1 2	250	1000,250,500,500,250,500
	1 4 1 1 3 1 1	230	230,920,230,230,690,230,230
	2 1 4 1 2 1 1	250	500,250,1000,250,500,250,250
Non beat[a]	1 3 2 3 2 1	270	270,972,378,972,378,270
	2 2 1 2 4 1	250	350,350,250,350,1000,250
	2 1 2 3 2 1 1	230	322,230,322,828,322,230,230
	3 2 2 1 1 1 2	250	900,350,350,250,250,250,350

[a]Note: For nonbeat rhythms, 2 = 1.4 and 3 = 3.6 interval ratios.

cortex (PMC), auditory cortex, inferior parietal lobe (IPL), basal ganglia, and cerebellum. We also examined these areas for evidence of beat sensitivity, which may have been obscured by regional averaging in previous univariate studies.

Materials and methods

Participants

A sample of 26 healthy young adults (14 female) participated in the study. Most participants reported music-playing experience (*M* = 6.54 years, *SD* = 5.20 years). Participants were excluded if they reported neurological impairment, use of psychotropic medication at the time of study, difficulty hearing, or if they did not meet general MRI environment safety criteria, such as metal in the body or a history of claustrophobia. All participants were compensated for their time monetarily, and procedures were approved by the Health Sciences Research Ethics Board at Western University.

Stimuli

Four auditory rhythms were generated for each of the 3 beat-strength conditions: strong-beat, weak-beat, and nonbeat rhythms. Twelve rhythms were taken from a larger set used in previous work (Grahn and Brett 2007; Grahn 2012); see Table 1 for a list of stimuli. Rhythms were composed of 500 Hz sine-wave tones. To aid perception in an acoustically noisy scanner environment, the tones "filled" most of the duration of each interval in the rhythms, ending 40 ms short of the specified duration to leave a 40 ms silent "gap" to demarcate the tone onsets (as in previous work, Grahn and Brett 2007). Thus, the interval durations for each rhythm were the tone interonset intervals. Each tone had 8 ms onset/offset ramps. Rhythms included 6 or 7 intervals and were about 3 s in duration. A final tone, the length of the shortest interval in the rhythm, was appended to the end of all sequences, such that the end of the final interval was demarcated by the final tone's onset.

Strong- and weak-beat rhythms were composed of integer-ratio intervals, such that interval durations were multiples of the shortest interval (i.e. 1:2:3:4). In nonbeat rhythms, the "2" and "3" intervals were replaced by 1.4 and 3.6 noninteger ratios (1:1.4:3.6:4), creating irregularity. Rhythms were presented at 3 different tempi, with the smallest interval being 230, 250, or 270 ms. Each beat-strength condition had 1 fast, 1 slow, and 2 medium tempo rhythms. Strong-beat rhythms contained interval arrangements that induced a perceptual accent at evenly spaced time points—at the beginning of each group of four units, marking the location of the beat (Povel and Essens 1985). Weak-beat rhythm intervals were arranged such that perceptual accents did not occur at evenly spaced time points, making any potential beat difficult to detect (Grahn and Brett 2007; Grahn 2012). Additionally, the noninteger ratios of the nonbeat rhythms eliminated regularity and necessarily led to irregularly spaced perceptual accents, such that no beat existed in the rhythm at all. During the fMRI scan, participants performed a discrimination task (comparing whether the third rhythm in a trial was the same as or different from the first two rhythms in that trial); therefore, a discrimination target rhythm was created for each stimulus by switching the order of the third and fourth intervals in the original rhythm.

Procedure

Rhythm discrimination task

While in the scanner, participants performed 8 blocks of the rhythm discrimination task. Each block lasted ~7 min. Within each block, every rhythm was presented on 2 trials, resulting in 24 total trials per block. Each trial included two presentations of a rhythm, with the screen depicting "First Listen" and "Second Listen" in text, followed by a presentation of the target rhythm. On half of the trials, the same rhythm, on the other half, a deviant rhythm in which the order of third and fourth intervals are switched. During the target rhythm, the screen displayed "Target Rhythm: Same or Different?". The target rhythm was followed by a 2-s response window with the text "Was the rhythm same or different? 1st finger = same, 2nd finger = different" in red letters on a black background. While this screen was displayed, participants responded via button press with either the index or middle finger of their right hand. Each rhythm was separated by a 1,300-ms interstimulus interval. Some trials were separated by an extended intertrial interval, unrelated to the task, to allow for silent baseline data collection. Stimuli were presented in a random order in every block. Before entering the scanner, participants were instructed how to complete the task and familiarized with the screens and response requirements. Participants were instructed to keep very still during the scan and specifically asked not to tap along with the rhythms.

Demographics questionnaire

After the scan, participants completed a demographics questionnaire that asked about age, gender, and musical experience.

Image acquisition and preprocessing

A Siemens MAGNETOM 7T MRI scanner was used to collect anatomical and functional images at the Center for Functional and Metabolic Mapping, Western University. An anatomical T1

image was collected after the first 4 functional runs. A head-only AC84 II gradient engine will be used with an 8-channel transmit/32-channel receive whole-head array. T2∗-weighted echo-planar imaging (EPI) data were collected using a 1-s TR, 22-ms TE, with a multiband acceleration factor of 3 and a 30° flip angle. EPI data were collected from 2 mm isotropic voxels in a 104 × 104 × 60 matrix.

SPM12 was used to preprocess the images. Functional and anatomical images were visually inspected and, when necessary, reoriented to be in a similar orientation. To account for subject movement, functional images were realigned to the mean functional image using second-degree B-spline interpolation. Each subject's functional images were then coregistered with their anatomical T1 image with a normalized mutual information cost function. T1 images were skull-stripped and segmented into gray-matter, white-matter, and cerebrospinal maps using the segment function and tissue probability maps included with SPM12. Multivariate analysis was performed on unsmoothed images in native space. For univariate analysis, coregistered images were spatially smoothed with an 8-mm full-width half-maximum kernel. Smoothed images were then spatially warped to Montreal Neurological Institute (MNI) template space before the first-level general linear model (GLM) was estimated.

First-level modeling

Univariate

A first-level GLM was implemented on the smoothed MNI-space functional images for each session. Independent regressors were entered for each beat strength condition (strong-, weak-, and nonbeat) and aligned to the onset of the first rhythm presentation and offset of the second presentation in each trial (thus covering the entire 2 presentations of a rhythm). The target rhythm (third presentation), button response, and 6 movement regressors were also entered into the first-level GLM. All trial regressors were modeled using an on–off boxcar method convolved with a canonical hemodynamic response function. Ultimately, the analysis resulted in one baseline-corrected (beta) image for each beat strength condition (modeling the first two presentations of a rhythm on a trial; target rhythms and responses were entered as nuisance regressors and not analyzed at the second level), for each subject. After this step, each subject's 3 baseline-corrected images (1 image per each beat strength condition) were used in the second-level analysis, including a one-way repeated-measures ANOVA using SPM12's flexible factorial model.

Multivariate: item-wise

The main RSA analysis was performed on item-wise activity patterns—one for each of the 12 rhythms. To calculate item-wise activity patterns, a first-level GLM was implemented independently from the univariate GLM. For each session, independent regressors were entered for each of the 12 rhythms and aligned to the onset of the first rhythm presentation and offset of the second presentation in each trial (thus covering the entire 2 presentations of a rhythm). As in the univariate GLM, the target rhythm, button response, and 6 movement regressors were also entered. All trial regressors were created using an on–off boxcar method convolved with a canonical hemodynamic response function. This analysis resulted in eight baseline-corrected (beta weight) images for each rhythm (one per run), for a total of 96 images.

Multivariate: condition-wise

The condition-wise GLM was identical to the univariate GLM, except that it was performed on unsmoothed native-space functional images.

Region of interest definition

The left and right pallidum, putamen, caudate, and SMA were the regions of interest (ROIs). Binary masks were created in MNI space using anatomically defined regions from the AAL3 atlas (Rolls et al. 2020). Masks were then warped from MNI space to each subject's native space using the SPM12 deformation tool and the inverse of the affine transformation matrix generated from the normalization step (see Image Acquisition and Preprocessing). All native-space ROIs were visually inspected for accuracy.

Whole-brain volumetric searchlight

In addition to ROI analyses, we examined activity patterns across the whole brain. A volumetric searchlight was employed to test pattern dissimilarities in overlapping small spherical ROIs. The volumetric searchlight considered data from small spheres of 160 voxels across each subject's whole brain in native space. Each voxel in the functional mask was used as a searchlight center. For each searchlight center, the nearest 160 voxels were considered for the activity pattern calculation. If a minimum of 160 voxels were not present in a sphere surrounding the searchlight center (e.g. on the edges of the functional mask), voxels nearest to the center (calculated by Euclidean distance, up to a maximum 30 mm radius) were added to the searchlight until ∼160 voxels were present.

For group-level analysis of the whole-brain searchlight, all averaging of the crossnobis distances (see Results) was performed in native space, creating whole-brain average crossnobis distance maps for each subject. Average crossnobis distances more than 4 times the interquartile range were removed from the whole-brain maps for each subject. The average crossnobis distance maps were then warped to MNI space for group-level testing. For the feature-encoding analysis (see below), a group-level map of possible beat-encoding regions was reverse-warped from MNI to native space to extract the full item-wise representational dissimilarity matrices (RDMs), which were averaged across searchlights contained in each cluster of the group-level map.

Representational similarity analysis

Univariate analyses average across fine-grained spatial patterns across voxels, thus only detecting overall increases or decreases in activity across a region. Multivariate pattern analysis quantifies patterns across multiple voxels and can detect item-specific activity patterns (Mur et al. 2009). We used RSA, a multivariate technique that tests for the encoding or representational specificity of regional activity by comparing the dissimilarity between multivariate activity patterns elicited by different stimuli (Kriegeskorte et al. 2008). Dissimilarity reflects sensitivity to differences in features between stimuli, such as the strength of the beat. We tested whether well-established beat-sensitive regions (i.e. the SMA and basal ganglia) encode individual rhythms regardless of beat (rhythm-encoding; Fig. 1B), or if activity patterns differ only according to beat strength (beat encoding; Fig. 1A), or if they showed both rhythm *and* beat encoding, in which activity patterns differ across individual rhythms, but with the greatest dissimilarity between rhythms of different beat strengths (Fig. 1C). To test for these different types of encoding patterns, we performed RSA in each ROI and across the whole brain (using a searchlight) on 12 activity patterns—one for each of the 12 rhythms.

Dissimilarity metric: cross-validated Mahalanobis distance estimate

Dissimilarities between activity patterns were quantified by the cross-validated Mahalanobis ("crossnobis") distance estimate

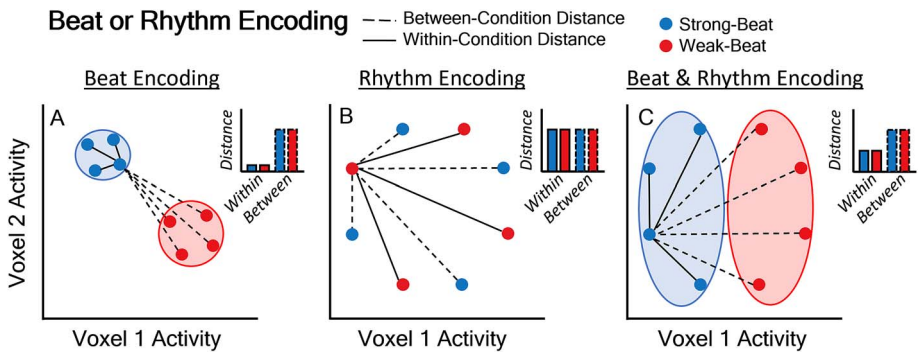

Fig. 1. Example activity pattern orientations in a 2-voxel ROI. Axes represent activity in each voxel. Small circles represent the 2-voxel activity patterns for individual rhythms; larger clouds show mean groupings. A–C) shows the theoretical layout for regions encoding beat A), individual rhythms B), or a combination of beat and rhythm C). Solid lines show the within-condition distances or the dissimilarity between rhythms of the same beat strength; dashed lines show the between-condition distances or the dissimilarity between rhythms of different beat strengths. Insets show the mean within-condition distances (left bars) and between-condition distances (right bars). In all 3 panels, significant between-condition distances would be found: Average distances between conditions would be greater than 0. However, true beat encoding is only present in A and C), as the between-condition distances are *greater* than the within-condition distances, on average. Differently, in B), the region activates in a unique pattern for each rhythm but does not distinguish on the basis of beat-strength—between-condition distances are equal to the within-condition distances.

(Diedrichsen et al. 2016). The distance estimate is similar to machine learning classifiers that classify items into groups based on the dissimilarity between activity patterns. However, the crossnobis distance returns a continuous measure of the dissimilarity between patterns, rather than a discretized classification accuracy, and is therefore informative about *how much* dissimilarity exists between activity patterns. To calculate the crossnobis distance, each stimulus was assigned a BOLD activity pattern consisting of beta weights from the first-level multivariate GLM (see Multivariate Modeling section below) for each voxel within an ROI or searchlight. For each of the 8 imaging runs, the activity patterns were multivariate noise normalized using the variance–covariance matrix of the first-level GLM residuals (Walther et al. 2016). Multivariate noise normalization downweighs the contribution of voxels relative to their residual variance (variance–covariance diagonal) and its correlational structure with other voxels included in the matrix (variance–covariance off diagonal). Because noise can be unique to each run, multivariate noise normalization was performed for each of the 8 functional runs independently.

Noise-normalized activity patterns were cross-validated using a leave-one-run-out approach. To cross-validate, distances between activity patterns were averaged across all but one imaging run. The distance was also calculated between activity patterns for the left-out run. Next, the inner product of the run-averaged distance and the left-out run distance vectors was calculated. The inner product reveals the consistency and magnitude of the two distance vectors, with 0 meaning no dissimilarity, and a possibility for negative distance if the vectors are in different spatial orientations—suggesting the dissimilarity is highly inconsistent across runs. The cross-validation steps were repeated by switching the left-out run on each iteration until all 8 runs were left out. The average of the 8 resulting inner products for each stimulus pair was taken as the cross-validated distance. Thus, each ROI and searchlight center exhibited 66 crossnobis distances per subject. Multivariate pattern analysis was performed in Matlab using the rsa toolbox (Nili et al. 2014), pcm toolbox (Diedrichsen et al. 2018), and custom code.

Summary statistic: categorically averaged Crossnobis distances

Pairwise dissimilarities were averaged into 3 within-condition categories, measuring the mean dissimilarity between rhythms of the same beat strength: (i) strong-beat vs. strong-beat (SB-SB); (ii) weak-beat vs. weak-beat (WB-WB); and (iii) nonbeat vs. nonbeat (NB-NB). Additionally, 3 between-condition categories were calculated, measuring the mean dissimilarity between rhythms of different beat strengths: 1. Strong-beat vs. weak-beat (SB-WB); 2. Strong-beat vs. nonbeat (SB-NB); and 3. Weak-beat vs. nonbeat (WB-NB).

Feature-encoding analysis: model testing

Although we categorized rhythms into three beat-strength conditions, individual rhythms also vary on features unrelated to beat. Neural patterns across rhythms may be partially driven by stimulus features unrelated to beat strength (e.g. tempo). Further, high-level features like beat strength may not be categorical, but continuous, with beat strength differing across stimuli even within one beat-strength condition. Activity patterns may therefore be sensitive to these continuous differences. To test whether more continuous models of beat strength, as well as other rhythmic features, explained the representational structure in our regions of interest, we created nine representational models relating to different rhythmic features and correlated the models with the neural item-wise representational dissimilarity matrices.

Nine models were created by assigning each of the 12 rhythms a vector representing the degree of the feature being modeled and then computing the squared Euclidean distance between the feature vectors for each rhythm (feature vectors and resulting representational models can be found in Supplemental Fig. 2). Thus, each value in a model RDM represents how dissimilar two rhythms are based on a given feature, identically to the neural RDMs representing the dissimilarity between activity patterns. The nine representational models were organized into three categories: basic feature models, condition-encoding models, and beat-strength models. Basic feature models included (i) the number of onsets, (iii) the number of different interval lengths (i.e. the number of 1-, 2-, 3-, and 4-duration intervals in each rhythm), and (iii) tempo. The condition-encoding models tested whether one beat-strength condition was represented uniquely from all other conditions. Thus, there was an independent encoding model for (iv) only strong-beat encoded, (v) only weak-beat encoded, and (vi) only nonbeat encoded. Finally, the beat-strength models tested whether the beat-strength representation was (vii) equal (rhythms of the same beat strength are not dissimilar, and are equally dissimilar to rhythms of different beat strength),

(viii) hierarchical (strong-beat rhythms are more dissimilar from nonbeat rhythms than from weak-beat rhythms), or (ix) sensitive to "counterevidence" using the beat counterevidence score (C-Score) model from Povel and Essens (1985). Counterevidence scores were calculated for each rhythm using a 4/4 meter. Unlike the hierarchical beat strength model, the C-Score model indicates the beat strength of each individual rhythm.

Using Pearson's *r*, the 9 models were correlated with neural RDMs for each subject and ROI. Thus, a distribution of 26 *r*-values was created for each model and ROI. To establish the expected correlation value of a successful model, we calculated the upper and lower bounds of the noise ceiling independently for each ROI. Noise ceilings estimate the range within which the true explanatory model (i.e. a perfect model containing all features encoded by a region) can be expected to relate to the data, given the intersubject variability in the sample (Lage-Castellanos et al. 2019). Thus, representational models that have correlations within the noise ceiling are likely to be an accurate estimate of the true representational structure of a region. Noise ceilings were created using a leave-one-subject-out approach (Nili et al. 2014). For the lower bound, each subject's neural RDM was correlated with the group-averaged RDM with that subject left out. The left-out subject was rotated until all subjects had been left out once. The lower bound of the noise ceiling is the average of each subject's correlation with the group. The noise-ceiling upper bound is calculated in the same way as the lower bound, except that the "left-out" subject is still included in the group-averaged RDM, increasing the correlation between each subject and the group RDM.

The feature-encoding analysis was performed on beat-sensitive a priori ROIs and on candidate feature-encoding regions from the exploratory whole-brain searchlight. To identify candidate feature-encoding regions, all searchlights with significant between-condition dissimilarities (compared to zero) were used in the feature-encoding analysis. Dissimilarities here may be driven by features that were balanced across beat-strength conditions (e.g. tempo), by features that change systematically across conditions (i.e. beat strength), or by a combination of features that differ between any set of individual rhythms. Therefore, comparisons in these regions are less controlled than the beat-encoding analyses and should be considered exploratory.

Results

Behavioral performance

Performance on the rhythm discrimination task was evaluated by calculating the percentage of correct trials for each beat strength condition. Overall, discrimination was significantly more accurate for strong-beat rhythms ($M = 78\%$, $SD = 15\%$), compared to weak-beat ($M = 68\%$, $SD = 12\%$), $t(24) = 3.99$, $P < 0.001$, and nonbeat rhythms ($M = 69\%$, $SD = 13\%$), $t(24) = 4.01$, $P < 0.001$. Discrimination performance for weak-beat and nonbeat rhythms was not significantly different, $t(24) = -0.44$, $P = 0.66$. Due to a technical error, responses from two subjects were not recorded, and they were removed from this analysis.

Univariate analysis

As a preliminary data check, whole-brain univariate analyses compared activation during rhythm listening versus rest and differences between beat strength conditions. An [all rhythms > rest] contrast revealed activation in the rhythm perception network, including bilateral SMA, PMC, basal ganglia, cerebellum, and superior temporal gyrus (STG) (Fig. 2). To test for univariate differences between beat strength conditions, a 1 × 3 (beat strength; strong, weak, non) repeated-measures ANOVA was performed at the whole-brain level. A main effect of beat strength revealed greater activity in the striatum for strong-beat rhythms compared to nonbeat rhythms (Fig. 2, right). The [strong-beat > weak-beat] and [weak-beat > nonbeat] contrasts did not reveal significantly different activation, and no other regions showed differences between beat strength conditions.

Multivariate pattern analysis

Multivariate activity patterns were analyzed in 8 a priori anatomically defined ROIs (left and right SMA, caudate, pallidum, and putamen). To determine whether activity patterns represented beat strength beyond that explained by individual rhythm encoding, the activity patterns from each rhythm (item-wise activity patterns) were extracted and tested for dissimilarity. Beat-encoding regions will have greater dissimilarity between activity patterns of *different* beat strength (e.g. [Strong vs. Non]) than those with the same beat strength (e.g. [Strong vs. Strong]), in which beat strength is identical (Fig. 1A and C). Thus, in each region, we tested whether the mean dissimilarities between rhythms of different beat strength conditions were greater than the mean dissimilarities within each beat strength condition combined (e.g. SB-WB > SB-SB + WB-WB).

Are individual rhythms, beat strength, or both encoded in motor region activity?

To examine this question, the crossnobis distances between each of the 12 rhythms (66 pairwise distances) were averaged into 6 categories of interest for statistical analysis—3 within-condition distances (SB-SB: strong vs. strong; WB-WB: weak vs. weak; and NB-NB: non vs. non), and 3 between-condition distances (SB-WB: strong vs. weak; SB-NB: strong vs. non; and WB-NB: weak vs. non). Planned *t*-tests indicated whether significant within- and/or between-condition distances were present ([SB-SB + WB-WB + NB-NB > 0] & [SB-WB + SB-NB + WB-NB > 0], respectively) and to test for greater encoding of beat strength than individual rhythms by comparing each mean between-condition distance to its constituent within-condition distances (e.g. [SB-WB > SB-SB + WB-WB]).

Group mean crossnobis distances for the six comparisons are shown in Fig. 3. To test for individual rhythm encoding while controlling for beat strength, the 3 within-condition distances were averaged and tested against 0 using a one-sample *t*-test. Because within-condition distances control for beat strength, regions with significant within-condition distances are likely to encode some other rhythmic feature, such as tempo or number of onsets, which may differ between rhythms of the same beat strength. Of the 8 ROIs, no regions had significant within-condition distances (all $Ps > 0.064$). To test whether beat sensitivity was detected in the item-wise activity patterns, the 3 between-condition mean distances were averaged and tested against 0 with a one-sample *t*-test. Of the 8 ROIs, 3 revealed significant between-condition distances: right putamen, $t(25) = 2.74$, $P = 0.01$; right caudate, $t(25) = 2.16$, $P = 0.041$; left SMA, $t(25) = 2.13$, $P = 0.04$. The SB-NB > SB-SB + NB-NB test revealed significantly greater distances between strong-beat and nonbeat conditions compared to within strong-beat and nonbeat conditions in 3 ROIs: left putamen, $t(25) = 2.57$, $P = 0.017$; left SMA $t(25) = 2.07$, $P = 0.049$; right SMA $t(25) = 2.80$, $P = 0.01$. All other $Ps > 0.11$. The SB-WB > SB-SB + WB-WB test revealed marginal differences in the left putamen, $t(25) = 1.77$, $P = 0.089$; and left caudate, $t(25) = 1.97$, $P = 0.061$. All other $Ps > 0.17$.

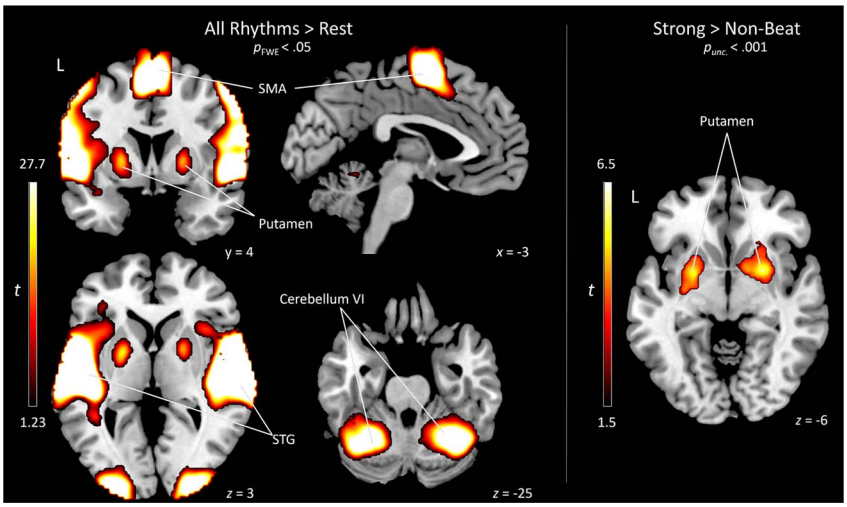

Fig. 2. Univariate activation for [all rhythms > rest] (left) and [strong-beat > nonbeat] (right) contrasts. Across all rhythms, the rhythm perception network was active, including left and right SMA, STG, putamen, PMC, and cerebellum VI. Strong-beat rhythms revealed greater activation in the putamen compared to nonbeat rhythms. The [strong-beat > weak-beat] and [weak-beat > nonbeat] contrasts did not show significant whole-brain differences.

Finally, the WB-NB > WB-WB + NB-NB test revealed only marginal differences in the left SMA, revealing greater within-conditions distances than between, $t(25) = -1.76$, $P = 0.09$. All other $Ps > 0.13$. Overall, this indicates that the left and right SMA and the left putamen encode beat strength: They show different voxel-wise activity patterns for rhythms with different beat strengths. The distances are greater between-condition than within-condition; thus, the effect is not caused by encoding *any* given rhythm with a unique activity pattern, but rather by more similar patterns for rhythms within a beat-strength category than between beat-strength categories.

Are mean activity patterns distinct between conditions?

To ensure that item-wise activity patterns are not only significantly different between conditions because of high variability in one condition (e.g. Supplemental 1A), we followed up the item-wise analysis by testing whether the condition-wise activity patterns for each of the 3 beat strength conditions were significantly dissimilar. Condition-wise activity patterns were derived from the first-level GLM in which all rhythms in a particular beat-strength category were modeled with the same regressor (the GLM that also gave rise to the univariate analyses), rather than individual rhythms being modeled with a unique regressor. The crossnobis distance between the mean activity patterns for each condition was calculated, resulting in 3 distances (strong vs. weak, strong vs. non, weak vs. non) for each ROI. Using a one-sample *t*-test (one-sample *t*-tests are suitable for crossnobis distances, as cross-validation centers the distribution of distances on 0; Diedrichsen et al. 2016), we found significantly dissimilar mean activity patterns between strong-beat and nonbeat rhythms in the left [$t(25) = 3.74$, $P = 0.001$] and right SMA [$t(25) = 3.20$, $P = 0.004$], and in the left [$t(25) = 2.90$, $P = 0.008$], and right [$t(25) = 2.32$, $P = 0.029$] putamen (Fig. 3, "Between" category on left bar graphs). This indicates that the SMA and putamen SB-NB differences found in the item-wise analysis were not driven by overlapping mean patterns with high item-wise variance. All other ROIs (caudate and pallidum) did not show significantly dissimilar mean activity patterns ($Ps > 0.06$). In all ROIs, strong- vs. weak-beat and weak- vs. nonbeat distances were not significant ($Ps > 0.061$), although the left caudate approached significance for strong- vs. weak-beat dissimilarity [$t(25) = 1.99$, $P = 0.057$]. Condition-wise dissimilarities for each ROI are shown in Supplemental Fig. 1.

How is beat represented in the SMA and putamen? Are basic rhythmic features represented?

The above analyses compared differences in beat strength based on categorically defined beat strength conditions (Grahn and Brett 2007). Both the item-wise and condition-wise tests revealed distinct activity patterns for strong-beat rhythms compared to nonbeat rhythms in the SMA and putamen. However, no region showed significantly dissimilar patterns between strong-beat and weak-beat or weak-beat and nonbeat conditions. When illustrated using multidimensional scaling (Fig. 4), it appears that activity patterns for weak-beat rhythms fall between strong-beat and nonbeat patterns—dissimilar from both but not enough to be significant when quantified by crossnobis distances. This could mean that beat is represented in these regions on a continuous scale, with item-level beat strength dictating activity patterns, rather than activity patterns representing the simple presence or absence of a beat. Separately, there may be variations in distance related to other features of the rhythms (e.g. tempo, number of onsets). Therefore, to test whether other features explained dissimilarities between rhythms, as well as to test other models of beat strength, we correlated 9 model RDMs with the neural RDMs extracted from the beat-sensitive regions—the left and right SMA and putamen (Fig. 5). Correlations between model and neural RDMs that reached the noise ceiling (see Materials and Methods) were considered good candidates for a true neural RDM. Models that do not reach the noise ceiling are considered insufficient in explaining the representational structure, suggesting that a better model exists (Yokoi et al. 2018). Representational model correlations were performed on a priori ROIs that showed beat sensitivity in previous analyses (i.e. the left and right putamen and SMA).

In both left and right SMA, the counterevidence score model (in which C-score indexes each rhythm's beat strength) was the winning model. It reached the noise ceiling, and it had the numerically largest correlation with neural RDMs. We compared

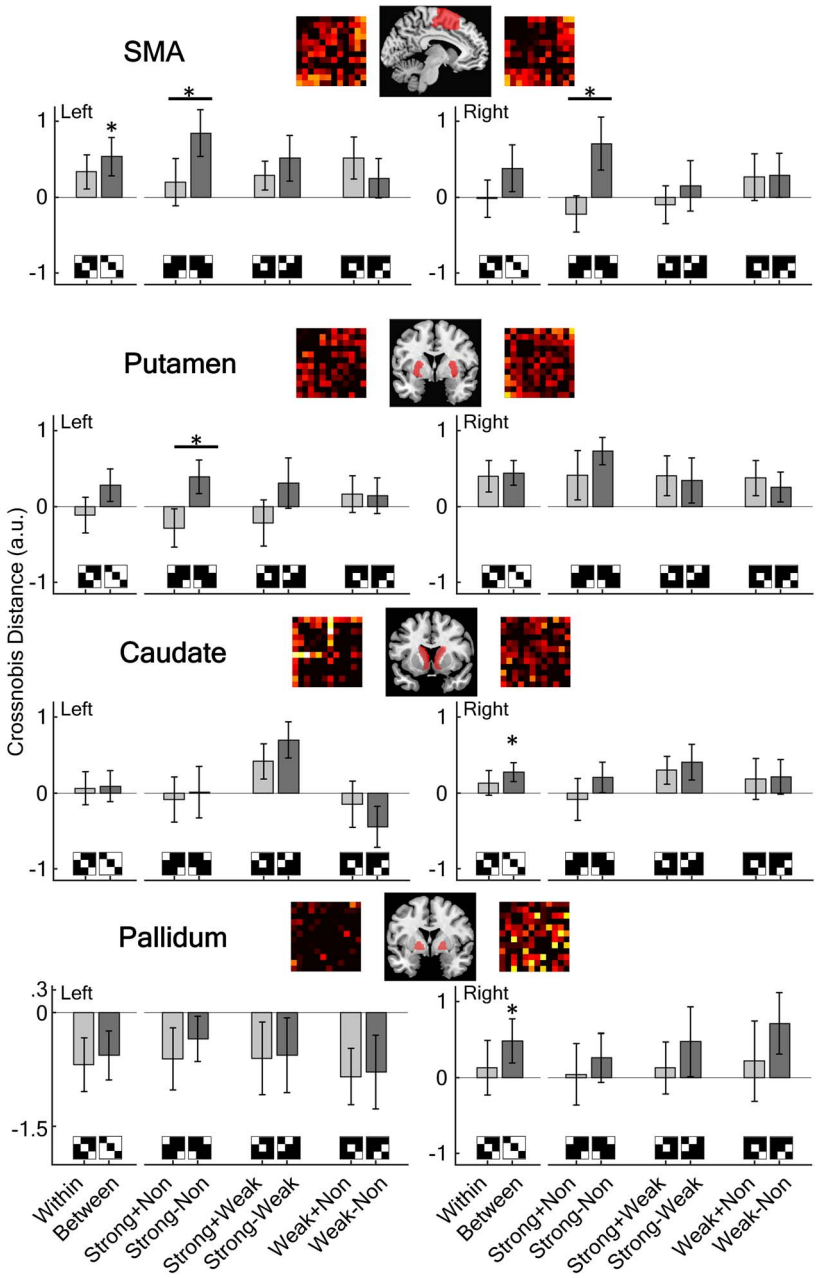

Fig. 3. Item-wise MVPA results in 8 ROIs. Bar graphs depict averaged within-condition distances (light gray) and between-condition distances (dark gray) in each ROI (panels). Inset: Representational dissimilarity matrices in each ROI for the 12 rhythms. Within- and between-condition distances were first averaged across beat strength conditions and tested against 0. For the two left-most bars in each panel, ∗ indicates where between or within-condition distances are greater than 0 at $P < 0.05$. For the other pairs of bars in the panel, between-condition distances (e.g. strong − non) were compared to the constituent within-condition distances (e.g. strong + non). Horizontal lines with ∗ indicate significantly greater dissimilarity for between-condition vs. within-condition distances at $P < 0.05$, consistent with beat strength encoding.

the winning model to the other 8 models using paired-sample *t*-tests on the distribution of Pearson's *r* values. In the left SMA, the C-Score model was significantly more correlated than the following models: Onsets [$t(25) = 2.76$, $P = 0.011$], Interval [$t(25) = 2.64$, $P = 0.014$], Tempo [$t(25) = 3.63$, $P = 0.001$], Weak Beat [$t(25) = 2.95$, $P = 0.007$], Nonbeat [$t(25) = 2.01$, $P = 0.056$], and Equal Beat [$t(25) = 2.16$, $P = 0.041$] models. The C-Score model was not significantly more correlated than the Strong Beat [$t(25) = 0.63$, $P = 0.53$] or Beat Hierarchy [$t(25) = 0.39$, $P = 0.70$] models. In the right SMA, the C-Score model was significantly more correlated than the Onsets [$t(25) = 3.72$, $P = 0.001$], Interval [$t(25) = 3.34$, $P = 0.003$], Tempo [$t(25) = 3.38$, $P = 0.002$], Strong Beat [$t(25) = 2.77$, $P = 0.010$], Weak Beat [$t(25) = 3.27$, $P = 0.003$], and Equal Beat [$t(25) = 2.56$, $P = 0.017$] models. The C-Score model was not significantly more correlated than the Nonbeat [$t(25) = 1.67$, $P = 0.11$] or Beat Hierarchy [$t(25) = 0.91$, $P = 0.37$] models in the right SMA. Thus, in left and right SMA, the C-Score model best explains the neural RDMs, suggesting that the SMA encodes beat strength on a rhythm-by-rhythm basis, beyond that explained by basic rhythm features or categorical beat strength.

Model RDM correlations were also calculated in the left and right putamen, which appeared beat-sensitive in previous analyses. Similar to the SMA, the C-Score model was the winning model in the left and right putamen (Fig. 5). However, because the noise

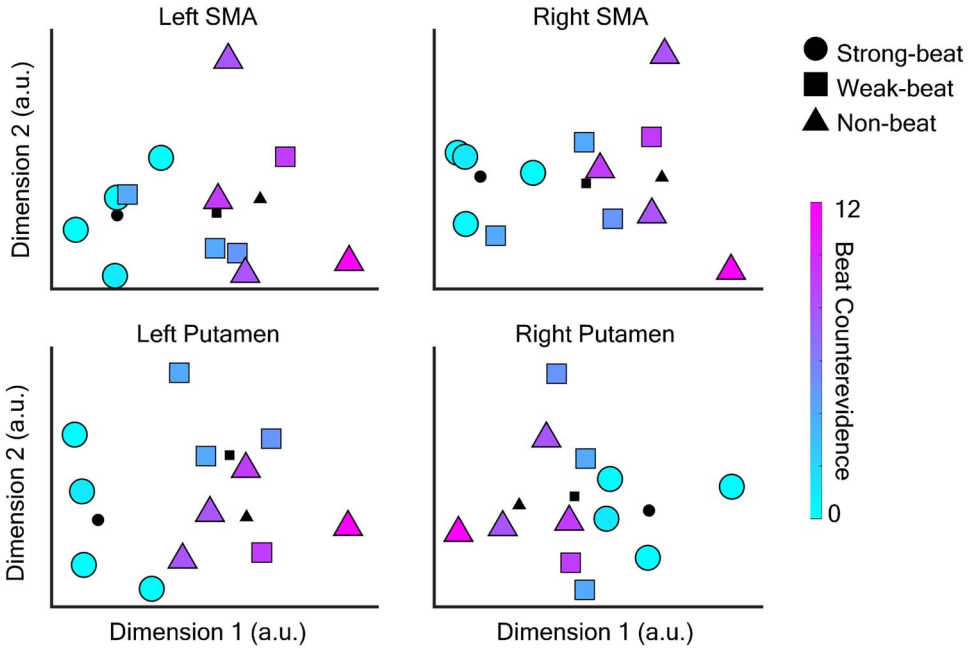

Fig. 4. Multidimensional scaling of activity patterns in left and right SMA (top) and putamen (bottom). Each colored data point represents the activity pattern for a rhythm. Small black data points represent the mean coordinates of each condition. Circles = strong-beat, squares = weak-beat, and triangles = nonbeat rhythms. Colors scale with the counterevidence score for each rhythm. Distances between data points represent the magnitude of dissimilarity between activity patterns. For visualization purposes only, each subject's negative distances were set to 0 before taking the group average RDM used for multidimensional scaling.

ceiling crosses $r = 0$ (suggesting high intersubject variability), we did not statistically test the strength of this model compared to others.

Exploring beat and rhythm encoding across the whole brain

In addition to testing a priori ROIs, we identified beat-encoding regions across the whole brain. Using a whole-brain searchlight, the item-wise activity pattern analysis was conducted in small spherical ROIs covering the brain. Identically to the a priori ROIs, distances from each searchlight across the brain were averaged to create the 6 categories described above (3 within-condition; 3 between-condition). The 6 comparisons of interest were entered into a second-level analysis using SPM12's flexible factorial with subject as a random effect. Whole-brain analyses followed the same order as the ROI analysis: We first tested for rhythm encoding while controlling for beat strength ([SB-SB + WB-WB + NB-NB > 0]) to identify regions that encode rhythm generally. Next, we tested for beat-encoding by identifying searchlights with greater between-condition dissimilarity than their constituent within-condition dissimilarities.

Whole-brain searchlight: rhythm encoding

To test for rhythm encoding, regardless of beat strength, we tested the averaged within-condition distances in each searchlight against zero [SB-SB + WB-WB + NB-NB > 0]. This test revealed rhythm encoding in left and right auditory cortices, right PMC, right cerebellum, and right occipital cortex (Supplemental Table 1).

Whole-brain searchlight: beat encoding

To identify beat-encoding regions across the brain, we tested whether between-condition dissimilarities were greater than within-condition dissimilarities using planned comparisons between each between-condition distance and the constituent within-condition distances, identically to the a priori ROI analysis. For the [SB-NB > SB-SB + NB-NB], only the SMA survived multiple comparison corrections at the cluster-size level ($P_{FWE} = 0.006$). This is consistent with findings in the anatomically defined SMA above. Finally, no differences were found between [WB-NB > WB-WB + NB-NB], or [SB-WB > SB-SB + WB-WB], even at a liberal threshold ($P_{unc.} < 0.001$).

Whole-brain searchlight: feature-encoding

To explore feature-encoding across the brain, we first identified candidate regions that may encode *any* of the rhythmic features by testing for regions with between-condition dissimilarities > 0. This identified regions sensitive to rhythm differences that could be further tested for sensitivity to basic features or other potential models of beat strength. That is, because basic features of the rhythms (e.g. tempo) were balanced across conditions, basic feature encoding regions *and* beat-encoding regions would both elicit between-condition distances (see inset in Fig. 1B and C). For example, a tempo-encoding region would have significant distances between slow and fast rhythms. Because the strong-beat and nonbeat conditions both have slow and fast rhythms, the mean between-condition distance (SB-NB) for tempo-encoding regions and beat-encoding regions would have significant between-condition distances. Thus, to explore encoding (basic feature and/or beat-encoding) across the whole brain, all searchlights with significant between-condition distances were passed on to the feature-encoding analysis.

As expected, many of the regions identified as having significant between-condition dissimilarity overlapped with rhythm-encoding regions identified in the within-condition test above, including bilateral auditory and PMCs, and right cerebellum (Supplemental Table 2). However, we additionally found between-condition dissimilarities in the bilateral SMA, IFGs, right putamen, and right pallidum, which were not shown in the within-condition test, and notably, only the SMA was identified as

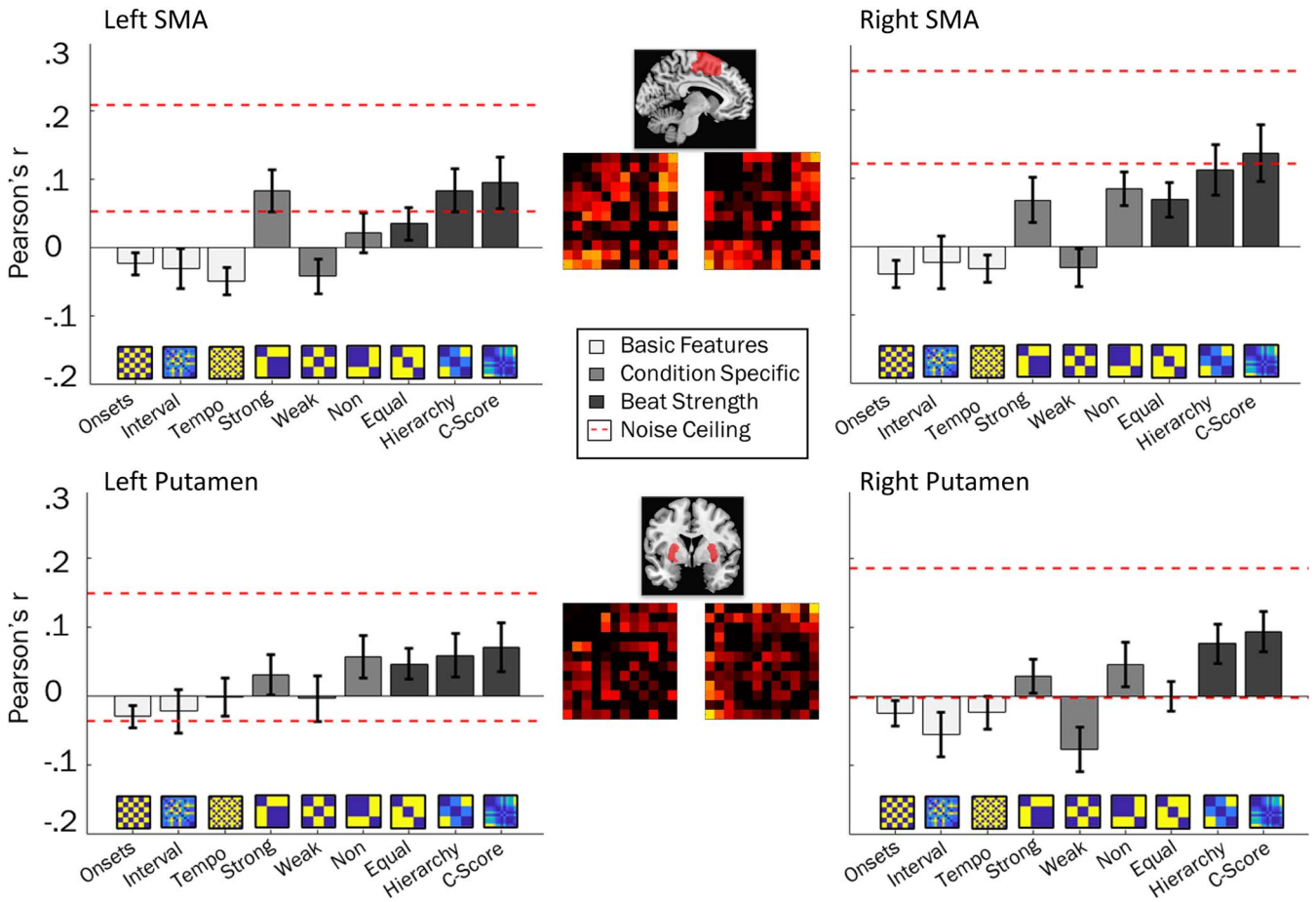

Fig. 5. MVPA results in anatomically defined SMA and putamen regions of interest. Red and black matrices are mean representational dissimilarity matrices for the *item-wise* activity patterns, thresholded at 0. Bar graphs show mean correlation values between each model RDM (*x* axis), and neural item-wise RDMs, error bars represent the standard error of the mean. Dashed lines indicate the upper and lower bound of noise ceilings: Models of good fit will breach the lower bound of the noise ceiling (Lage-Castellanos et al. 2019).

beat-encoding as reported in the previous section. This suggests that while the SMA is beat sensitive, the remaining regions identified in this exploratory analysis appear to have a more nuanced representation—when beat strength is controlled, dissimilarities are relatively weak (within-condition is not greater than 0), but when there are differences in beat strength, the patterns begin to differ more reliably (between-condition dissimilarities are greater than 0) but are not significantly greater than within-condition distances. This indicates that multiple features may be represented in the activity patterns from these regions.

To determine which features are represented in activity patterns across the whole brain, the significant between-condition searchlights were passed on to the feature-encoding modeling analysis as performed on the a priori ROIs. For each of the 12 clusters in the [SB-NB + SB-WB + WB-NB > 0] test (Fig. 6; Supplemental Table 2), the 9 representational models were correlated with each subject's RDM, averaged across all searchlights contained in each cluster. Because clusters 1, 2, and 3 clearly crossed functional and anatomical boundaries, these clusters were further divided using the AAL parcellation. Regions were not reported if the noise ceiling crossed 0 (the data are too noisy to reliably fit a model), or if none of the mean model correlations breached the lower bound of the noise ceiling (no model sufficiently explains the data). Interpretation of feature encoding is restricted to regions with evidence for rhythm and/or auditory encoding from past literature. This process resulted in 20 regions with successful model fits. To reduce the number of tests, in each region, we only statistically tested the model correlations between models breaching the lower bound of the noise ceiling. Results of the model correlations for the whole-brain clusters can be found in Table 2 and Fig. 6.

In motor areas, significant searchlights were found in the left and right PMCs (precentral gyri and left postcentral gyrus) and in the SMA. In the left PMC (precentral gyrus), the Strong Beat, Nonbeat, Beat Hierarchy, and C-Score models breached the lower bound of the noise ceiling. Paired sample *t*-tests revealed no differences between these models ($Ps > 0.053$). In the left postcentral gyrus, only the C-Score model breached the noise ceiling. In the right PMC, (precentral gyrus), the Beat Hierarchy and C-Score models breached the noise ceiling. Comparisons between these models revealed no significant difference ($P = 0.68$). In the SMA, similar to the a priori SMA ROI, the Beat Hierarchy and C-score models breached the noise ceiling but were not significantly different from each other ($P = 0.60$).

The right IFG was parcellated into the operculum, triangularis, and orbital cortex. In the right inferior frontal operculum, the Strong Beat, Nonbeat, Beat Hierarchy, and C-Score models all breached the noise ceiling. Comparisons between these models revealed no significant differences ($Ps > 0.14$). In the right inferior

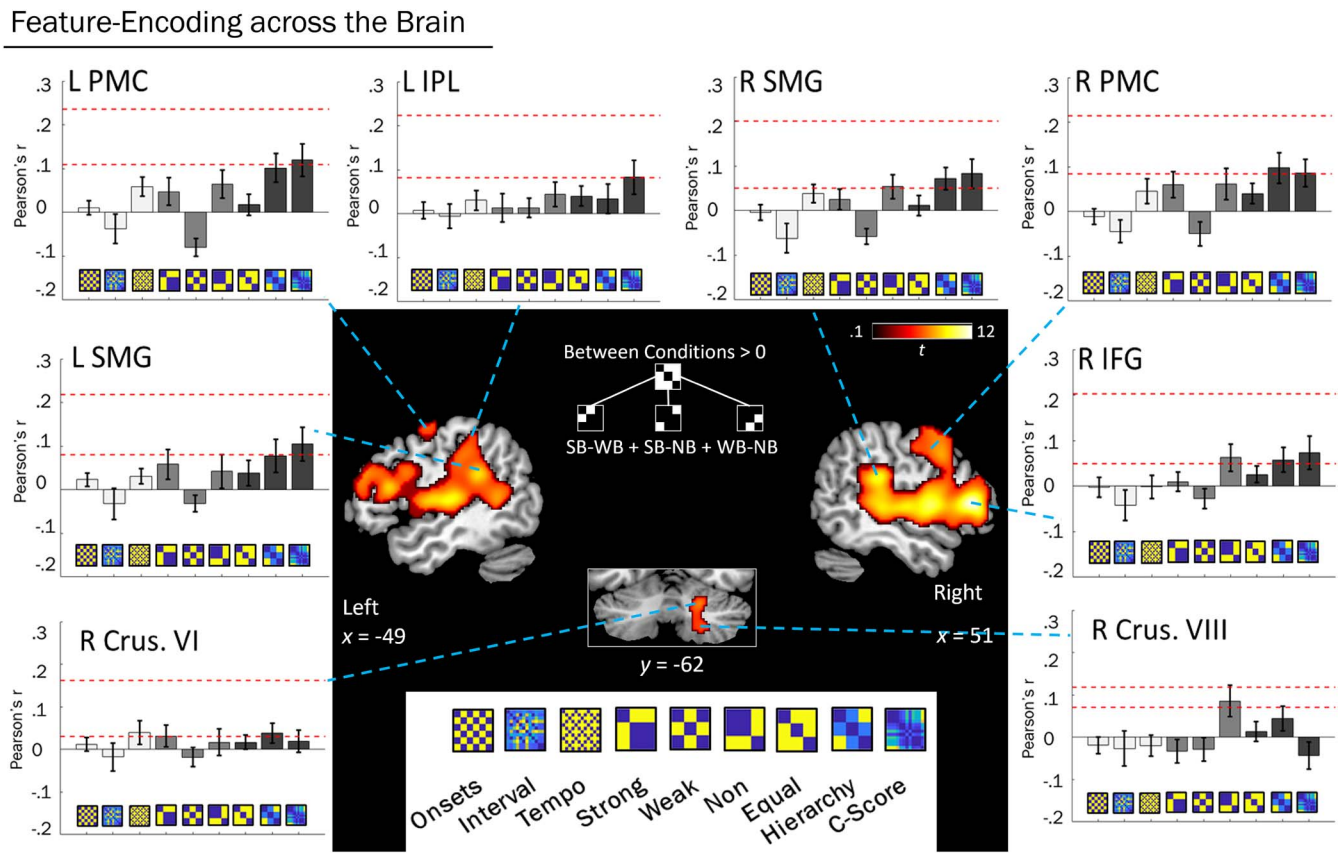

Fig. 6. Results of the feature-encoding model tests on clusters with significant [between condition > 0] searchlights. Heat map indicates the strength of the *t*-test [SB-WB + SB-NB + WB-NB > 0], indicating areas that encode rhythmic features. Insets: Group mean and standard error of Pearson correlations for each representational model in each brain region. White, light-gray, and dark-gray bars represent basic feature, condition-specific, and beat strength models, respectively. Left and right SMGs, PMC, IPL, and IFG were most correlated with models of beat strength. Cerebellum lobule VI ($*n = 25$) was most correlated with the tempo, strong-beat, and beat hierarchy models. Cerebellum lobule VIII ($*n = 19$) was most correlated with the nonbeat model. Right temporal pole and right medial temporal gyrus model fits not shown.

Table 2. Whole brain searchlight: feature-encoding results.

Region	Cluster	Models above noise ceiling lower bound	Best fitting model	Mean fit (M_r)	SD
L precentral gyrus	9	Strong-beat, Nonbeat, Beat Hierarchy, C-Score	Beat Hierarchy	0.10	0.15
SMA	6	Beat Hierarchy, C-Score	C-Score	0.11	0.19
R parahippocampal gyrus	8	Onsets, Nonbeat	Onsets	0.048	0.12
R inferior frontal operculum	1	Strong-beat, Nonbeat, Beat Hierarchy, C-Score	Beat Hierarchy	0.085	0.14
R inferior frontal triangularis	1	Nonbeat, Beat Hierarchy, C-Score	C-Score	0.073	0.19
R inferior frontal orbital cortex	1	Strong-beat, Beat Hierarchy, C-Score	C-Score	0.086	0.13
R precentral gyrus	1	Beat Hierarchy, C-Score	Beat Hierarchy	0.098	0.18
R insular cortex	1	Tempo, Beat Hierarchy, C-Score	C-Score	0.067	0.13
R superior temporal pole	1	Onsets, Tempo, Strong-beat, Beat Hierarchy, C-Score	Tempo	0.061	0.15
R middle temporal gyrus[a]	1	Strong-beat, Beat Hierarchy, C-Score	C-Score	0.096	0.13
R supramarginal cortex	1	Nonbeat, Beat Hierarchy, C-Score	C-Score	0.083	0.17
L inferior frontal triangularis	2	Nonbeat, C-Score	C-Score	0.078	0.22
L inferior frontal operculum	2	C-Score	C-Score	0.070	0.20
L postcentral gyrus	2	C-Score	C-Score	0.12	0.19
L inferior parietal lobe	2	C-Score	C-Score	0.083	0.20
L supramarginal gyrus	2	C-Score	C-Score	0.10	0.20
R fusiform gyrus	3	Nonbeat	Nonbeat	0.047	0.13
R cerebellum lobule VI[a]	3	Tempo, Strong-beat, Beat Hierarchy	Tempo	0.039	0.14
R cerebellum lobule VIII[a]	3	Nonbeat	Nonbeat	0.085	0.16
R primary visual cortex	7	Nonbeat	Nonbeat	0.029	0.16

Note: Regions with a noise ceiling crossing 0, or with no model breaching the lower bound of the noise ceiling are not reported. [a]Regions with missing data: Middle temporal gyrus ($n = 24$), cerebellum lobule VI ($n = 25$), and cerebellum lobule VIII ($n = 19$).

frontal triangularis, Nonbeat, Beat Hierarchy, and C-Score models all breached the noise ceiling. Comparisons between models revealed no significant differences between the mean correlations of these three models ($Ps > 0.58$). In the right inferior frontal orbital cortex, Strong Beat, Beat Hierarchy, and C-Score models all breached the noise ceiling. Comparisons between models revealed that the mean correlation of the C-Score model was marginally greater than the Strong Beat model [$t(25) = 1.81$, $P = 0.08$]. No other significant differences were detected ($Ps > 0.11$). In the left hemisphere, the IFG only covered the triangularis and operculum. In the left inferior frontal triangularis, the Nonbeat and C-Score models breached the noise ceiling. Comparisons between models revealed no significant differences between models ($P = 0.87$). In the left inferior frontal operculum, only the C-Score model breached the noise ceiling. Overall, this suggests that representations of rhythm in the right and left IFGs relate to beat strength, possibly on the individual rhythm level. In the right insular cortex, Tempo, Beat Hierarchy, and C-Score models breached the noise ceiling. Comparisons between these models revealed no significant differences between models ($Ps > 0.56$).

The temporal lobe had significant searchlights throughout auditory cortices, but many did not have a model of good fit. However, the right superior temporal pole and the right middle temporal gyrus were correlated with feature-encoding models. In the right superior temporal pole, the Onsets, Tempo, Strong Beat, Beat Hierarchy, and C-Score models breached the noise ceiling. Comparisons across these models revealed no significant differences ($Ps > 0.28$). In the right middle temporal gyrus, the Strong Beat, Beat Hierarchy, and C-Score models breached the noise ceiling. Comparisons across these models revealed no significant differences between models ($Ps > 0.13$). In the right parahippocampal gyrus, the Onsets model and Nonbeat model breached the noise ceiling but were not significantly different [$t(25) = 0.50$, $P = 0.62$].

In the left IPL, dorsal to the supramarginal gyrus (SMG), and in the left SMG, only the C-Score model breached the noise ceiling, suggesting that the IPL encodes beat strength at the individual rhythm level. In the right SMG, the Nonbeat, Beat Hierarchy, and C-Score models breached the noise ceiling. Comparisons across these models revealed no significant differences between models ($Ps > 0.32$).

In the right cerebellum, lobules VI and VIII had significant searchlights. In the right lobule VI, the Tempo, Strong Beat, and Beat Hierarchy models breached the noise ceiling. Comparisons across these models revealed no significant differences between model fits ($Ps > 0.79$). In right lobule VIII, only the Nonbeat model breached the noise ceiling. This suggests that representations of rhythm in lobule VIII relates to differences between nonbeat rhythms and all other rhythms.

In the right primary visual cortex and right fusiform gyrus, only the Nonbeat model breached the noise ceiling. As the visual cortex is not known to be sensitive to rhythm, we have not interpreted this result.

Discussion

In this experiment, we examined brain areas that encoded rhythmic properties by testing the dissimilarity between fine-grained activity patterns in previously identified beat-sensitive regions, as well as across the whole brain. We tested a priori anatomically defined ROIs for the left and right SMA, caudate, pallidum, and putamen. Of these, the bilateral SMA and bilateral putamen activated in distinct activity patterns for rhythms with different beat strengths. Moreover, SMA and putamen activity patterns also encoded the relative differences in beat strength between individual rhythms—the magnitude of pattern dissimilarity corresponded to the difference in beat strength for each rhythm. This suggests that the activity patterns did not simply reflect the presence or absence of a beat; they also reflected evidence degree of beat strength. We did not find evidence that other basal ganglia structures (i.e. the caudate and pallidum), encoded rhythmic features or were sensitive to beat strength, though right caudate pattern dissimilarities between beat strength conditions were marginally significant. Thus, building on previous univariate findings (Grahn and Brett 2007; Chen et al. 2008a, 2008b), this study is the first to show that the SMA and putamen encode beat strength via multivoxel activity patterns and that these patterns are dictated by the strength of the beat for a given rhythm. Our data also clarify that the SMA and putamen do not encode each individual rhythm with equal discriminability (as would be predicted if individual interval durations were driving activity patterns); rather, beat strength is the driving feature behind SMA and putamen activity patterns during rhythm listening, with more distinct patterns between rhythms of different beat strengths, compared to rhythms of similar beat strengths.

In addition to a priori ROIs for the SMA and basal ganglia, rhythm- and beat-encoding regions across the brain were identified using a whole-brain searchlight. Identically to the ROI analyses, the searchlights were analyzed to find regions that encoded beat strength: Activity pattern dissimilarities for rhythms of different conditions were compared to dissimilarities within conditions. After controlling for within-condition dissimilarities, we found regions encoding differences in beat strength beyond that explained by individual rhythm encoding. The only beat-encoding region identified in whole-brain analysis was the bilateral SMA, as also shown in the a priori ROI analysis.

We also explored individual rhythm encoding across the whole brain to reveal areas that were sensitive to differences between rhythms, regardless of beat strength. Pattern dissimilarities between rhythms of the same beat strength were detected in the bilateral PMC and IFGs, temporal pole, superior temporal gyrus, medial temporal lobe, medial temporal gyrus, and cerebellum. These regions align well with the univariate activation found both here (in the all Rhythms > rest contrast), and in previous work (Schubotz et al. 2000; Chen et al. 2006, 2008b; Chen et al. 2008a; Grahn and Brett 2007; Bengtsson et al. 2009; Grahn and McAuley 2009; Grahn and Rowe 2009; Grahn and Schuit 2012; Kung et al. 2013; Lerens et al. 2013; Matthews et al. 2020; Hoddinott et al. 2021).

Finally, we explored rhythmic feature representations across the whole brain by fitting representational models to regions with significant between-condition dissimilarity, which may have reflected beat, rhythm, or basic feature encoding. The between-condition dissimilarity was used to identify candidate regions in which representations may relate to *any* rhythmic feature (beat strength, basic features). Beat strength was largely represented across the rhythm perception network—left and right PMCs, SMA, IFG, and SMG correlated best with beat strength models. Importantly, of these regions, only the SMA had activity patterns driven mainly by beat strength, as this was the only region to have greater dissimilarity between beat strength conditions, compared to within conditions, an analysis in which all beat-unrelated rhythmic features are well controlled. In essence, this means SMA activity patterns are most distinct between rhythms of different beat strengths, as depicted in Fig. 1A and C, while IFG and SMG

activity patterns are partially driven by features or processes other than beat, such as in Fig. 1B and C.

In the temporal lobe, many regions had no winning model. Early auditory cortical regions, such as Heschl's gyrus, may encode each sound with a unique activity pattern and may not distinguish rhythms along the features for which we made distinct models, alternatively, Heschl's gyrus may encode many or all basic features of the rhythms, which would result in highly dissimilar activity patterns between all rhythms, but dampen the explanatory impact of any one basic feature model. Activity patterns in the right superior temporal pole correlated with multiple models, including the Onsets, Tempo, Strong-beat, Beat Hierarchy, and C-Score models, though numerically, the tempo model fits best. This is perhaps indicative of auditory regions encoding many basic and nonbasic stimulus features. Similarly, activity in the right insula was correlated with the Tempo, Beat Hierarchy, and C-Score models, though C-Scores were the model of best fit and may represent multiple features of rhythm.

Overall, the current study identified the SMA and putamen as beat-encoding regions, and activity in the right PMC, IFGs, SFGs, and left IPL may partially relate to beat strength. Each of the identified regions fit well with previous univariate work in rhythm and beat perception. The SMA and putamen are consistently reported to activate more during strong-beat rhythms, compared to non-beat rhythms (Grahn and Brett 2007; Chen et al. 2008b; Chen et al. 2008a; Bengtsson et al. 2009; Grahn and Rowe 2009, 2013; Chapin et al. 2010; Matthews et al. 2020; Hoddinott et al. 2021). In addition, many previous studies report a larger rhythm perception network incorporating the remaining regions identified in our exploratory analysis: Left and right auditory cortices, SMA, basal ganglia, PMCs, and cerebellum all activate during rhythm listening but do not necessarily show sensitivity to the beat (Grahn and Brett 2007; Chen et al. 2008a, 2008b). These findings were replicated with our univariate analysis, except for the effect of beat strength in the SMA—no differences were found univariately between beat strength conditions; however, our data revealed multivariate encoding of beat strength in the SMA. Notably, null effects are somewhat difficult to interpret through the lens of previous univariate work—as the current study used a more repetitive task design (only four rhythms per condition) and an MRI scanner with greater field strength (7T vs. 3T), which can impact signal-to-noise ratios (Triantafyllou et al. 2005). Further, previous work did not find univariate SMA differences between beat strength conditions at the whole-brain level but only in ROI analyses (Grahn and Brett 2007).

SMA activity may reflect the predictability of rhythms. As the participants hear the rhythms repeatedly across the experiment, the rhythms may be learned by the later imaging runs. During debriefing, nearly all subjects reported that the rhythms were repeated. Thus, the lack of a univariate effect of beat in the SMA may reflect the equalization of predictability between rhythms of different beat strengths. While the strong-beat rhythms are somewhat predictable (partly because of their regularity and partly because we experience similar rhythms in real music) at the beginning of the experiment, the weak- and nonbeat rhythms are relatively unpredictable. But, as familiarity increases with exposure, the prediction-related activity in the SMA may become more similar between conditions. Unfortunately, in the current experiment, this change in predictability correlates with the amount of time in the scanner, making inferences about early vs. late contrasts confounded by physiological run-to-run noise, and neural habituation as described above.

The inferior frontal cortex activity patterns had significant between-condition distances and correlated with beat strength models. This finding aligns well with previous univariate work that reported greater activation in overlapping coordinates in the IFG for strong-beat rhythms compared to weak- and non-beat rhythms (Grahn and Brett 2007), perhaps revealing a beat-sensitive region that has not been thoroughly explored in rhythm perception. Outside of rhythm perception work, studies using naturalistic music have shown that the IFG is more active when people listen to music, compared to amusical control conditions (Levitin and Menon 2003), a contrast that may also reveal areas active for beat vs. no-beat stimuli. The IFG also activates when there is rhythmic tension in music, and when subjects are tapping a steady beat during a rhythm with high counterevidence for that beat (Vuust et al. 2006, 2011). Rhythmic tension fits well with the current finding that activity patterns in the IFG correlate with beat strength.

Representations of rhythm in the left and right IPLs were most correlated with the beat strength counterevidence score model, compared to other models. The left and right SMGs had significant between-condition distances, and neural representations were most correlated with models of beat strength. The SMG has been reported active for rhythm perception compared to silence, and for highly complex compared to simpler rhythms (Kasdan et al. 2022). One study reported that SMG activity is lateralized, with the left SMG responding to changes in pitch, and the right SMG responding to rhythm (Schaal et al. 2017), though there is also evidence that musicians recruit the right SMG for pitch recognition tasks (Schaal et al. 2015). Here, we did not manipulate pitch, and we found that SMG encoding was related to beat strength in both hemispheres. The SMG may encode in sequential order. One transcranial magnetic stimulation (TMS) study reported that downregulation of the left SMG increased participants' reporting of memorized verbal and visual sequences in the incorrect order, while the content of sequences (e.g. letters in a memorized string) remained stable (Guidali et al. 2019). This suggests that the SMG is necessary to encode the order of sequential stimuli, such as a list of letters, but not necessarily the content—which letters were in the list. However, if the SMG *only* encodes sequential order, one would expect similar encoding of all rhythms in the current study—each rhythm had a unique order of intervals. Instead, we found that left and right SMGs correlated most with counterevidence score models of beat perception.

Though beat sensitivity was found bilaterally in the SMG, in the left hemisphere only, the between-conditions cluster extended dorsally from the SMG, crossing the intraparietal sulcus and then medially into the IPL. Similar to the SMG, representations of rhythm in the ventral left IPL were most correlated with the beat-strength counterevidence score model. Theoretical models of beat perception predict the IPL to play a role (Patel and Iversen 2014). Specifically, the parietal lobe's position in the dorsal auditory stream makes it a good candidate for crosstalk between auditory and motor regions. It has been proposed that the parietal lobe encodes rhythmic information, and may facilitate motor predictions of upcoming acoustic events, as predicted by the Action Simulation for Auditory Prediction (ASAP) hypothesis (Patel and Iversen 2014), as well as recent motor physiological explanations for the role of the motor system in beat perception (Cannon and Patel 2021). Empirical evidence also suggests a role for the parietal cortex in beat perception—down-regulation of the left posterior parietal cortex with TMS disrupts beat-based timing but not absolute interval timing, suggesting that the left parietal cortex

plays a role in beat-related behavior (Ross et al. 2018). Because the current task did not require explicitly timed motor responses, our findings expand on previous brain stimulation experiments by showing that parietal cortex activity correlates with beat strength in rhythms during perception without action.

The regions we found that encoded rhythm, but were not sensitive to the beat, also align with previous research. The cerebellum is commonly found to be active in rhythm and timing-related tasks (Lee et al. 2007; Coull and Nobre 2008) and may be involved in absolute time perception, encoding the absolute duration of intervals (Grube et al. 2010; Teki et al. 2011) as opposed to the relative duration between two intervals (e.g. encoding intervals in a rhythm relative to the underlying beat interval). Indeed, TMS of the cerebellum affects the estimation of single-interval durations in tasks requiring absolute interval encoding (Lee et al. 2007). Our finding that the cerebellum lobule VI represents rhythms according to their overall tempo aligns well with an absolute timing process: Rhythms of different tempi must also have intervals of different absolute duration. It then makes sense that cerebellar activity patterns are highly dissimilar from each other when rhythms are made up of intervals that have different absolute durations. In cerebellum lobule VIII, we found that activity patterns differentiated nonbeat rhythms from all other rhythms (the Nonbeat condition-specific model). This may also be explained by absolute timing in the cerebellum—the nonbeat condition was the only condition in which rhythms included noninteger ratios. These noninteger ratios are impossible to encode relative to other durations, such as a beat or by subdividing longer intervals in the same rhythm. Thus, to encode nonbeat rhythms, absolute timing must be used. Cerebellum lobule VIII correlated most with the Nonbeat condition model, which may indicate activation for rhythms that require absolute timing strategies.

Many searchlights throughout the auditory cortex represented individual rhythms. Bilateral STG, Heschl's gyrus, temporal poles, and other auditory regions revealed significantly dissimilar activity patterns across rhythms. However, we were not able to create a sufficient model to explain which features of the rhythms were represented in the auditory cortex. Auditory areas may encode many features of the rhythms and sounds, so many that we cannot create a single model to capture this representation. Notably, the noise ceiling in the auditory cortex is quite high and does not show large variability between subjects—the upper and lower bounds are relatively narrow. This suggests that even though our models do not sufficiently explain the representations of rhythm in the auditory cortex, activity patterns are quite reliable across subjects. More work is needed to reveal how rhythmic acoustic stimuli are represented in the auditory cortex. Future studies may consider using larger, feature-rich stimulus sets, as opposed to the highly controlled rhythms used in the current experiment. For example, some work has been performed using naturalistic music and speech (Norman-Haignere et al. 2015), but more can be done to specifically target beat strength using naturalistic music.

The neural representation of rhythm begins in the auditory system and transforms into beat-strength representation in motor and association regions. In general, the current findings align with the theoretical role of the dorsal auditory stream in beat perception (Patel and Iversen 2014). Specifically, auditory information is proposed to pass dorsally from the auditory cortex, through the inferior parietal cortex, and into the PMC (i.e. the SMA). Motor-based predictions in the SMA are sent back through the dorsal stream to predict incoming auditory information. Simultaneously, auditory information is sent from the auditory cortices to the inferior frontal cortex via the ventral stream. This theory would predict that sensory regions encode rhythms generally, without extra sensitivity to beat strength, while premotor regions should have the strongest representation of beat strength. Further, the IPL (dorsal stream) and IFG (ventral stream), which may be involved in integrating incoming auditory information and top–down predictions from the PMC, may reflect both bottom–up sensory representation of the rhythms and top–down beat strength representation. Indeed, our current data show that in the auditory cortex, activity patterns for each rhythm are highly dissimilar but not explained by low- or high-level features of the rhythms. These sensory regions are likely to respond to each individual sound in the rhythm. As the sensory information is passed to association regions, the beat becomes the prominently encoded feature. The putamen, SMA, IPL and SMG, and the IFG appear to represent rhythms based on their individual beat strength. This explanation fits with the nuanced representation of rhythm in these regions—the IPL and IFG did not have significant within-condition dissimilarities, indicating that they do not represent each individual rhythm, as auditory sensory regions do, but they also did not have greater between-condition dissimilarities compared to within, suggesting that they do not represent beat above and beyond individual rhythms, as do the SMA and putamen. Thus, these association regions may reflect contributions of beat-encoding and rhythm-encoding activity that is passed down from the SMA and auditory cortex simultaneously. However, because the temporal resolution of fMRI is very slow, it is not possible to show which of these regions first represents the beat. However, compared to the other regions, the SMA, and possibly the left putamen, appear to be the *most* sensitive to the beat—as these are the only regions to have greater dissimilarities between beat strength conditions compared to within beat-strength conditions. This was especially the case for the SMA, which is the only region to survive multiple comparison corrections in the [SB-NB > SB-SB + NB–NB] contrast at the whole-brain level and is corroborated by past univariate work.

The putamen's role in beat perception also aligns with recent theoretical work that expands on the ASAP hypothesis (Cannon and Patel 2021). The striatum may respond to regular auditory stimulation by signaling a chunk of sequential predictions to be carried out by the SMA (Cannon and Patel 2021). The SMA compares the prediction to the incoming auditory stimulation and reports some form of prediction error back to the striatum to update the following prediction. This would align well with our current findings in both regions: Rhythms with high counterevidence for the beat are relatively unpredictable. Thus, as counterevidence increases, unpredictability, and prediction error, also increases, resulting in highly distinct activation patterns. When there is a strong, highly predictable beat, there is little prediction error, and signaling from the SMA is very similar across rhythms.

In conclusion, this experiment is the first to show that the SMA and putamen activate in highly distinguishable multi-voxel patterns for rhythms of different beat strengths. Importantly, the beat was represented on a continuous scale—the greater the difference in beat strength between two rhythms, the more dissimilar the activity patterns were for each of the rhythms. Through exploratory analyses, we also show that beat strength is represented, at least partially, in the left inferior parietal lobe and the right PMC. Lastly, cerebellum activity patterns were altered most by tempo, suggesting that the cerebellum may be tuned to the overall rate of rhythms.

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Author contributions

Joshua Hoddinott (Data curation, Formal analysis, Investigation, Methodology, Visualization, Writing—original draft, Writing—review & editing) and Jessica Grahn (Conceptualization, Funding acquisition, Project administration, Resources, Supervision, Writing—review & editing).

Supplementary material

Supplementary material is available at *Cerebral Cortex* online.

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